

When the Faithfulness Metric Fails Before the Explanation Does: Layer-wise Relevance Propagation on a Self-Supervised CT Foundation Model

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Abstract—Attribution maps make a testable claim: the voxels they rank highest are the ones the model used. Deletion metrics test that claim by masking top-ranked voxels and watching the prediction fall. We apply this protocol to a self-supervised CT foundation model, CT-FM, fine-tuned for lung-nodule malignancy on volumetric chest CT, comparing three layer-wise relevance propagation composites against Grad-CAM. Two findings follow. First, the pretraining is worth having: CT-FM initialisation improves test AUC over an architecturally identical randomly-initialised network at every label budget, reaching 0.895 against 0.873 at full supervision, though a seed replicate shows the smallest gain lies within seed noise. Second, and more consequentially, *no attribution method outperformed a random-order deletion baseline*. All four scored worse, by 0.12 to 0.22 AOPC. We trace this to the baseline rather than the explanations: random deletion scatters replaced voxels throughout the volume and drives the input off the data manifold, whereas attribution-guided deletion removes a compact contiguous region that remains far more plausible as a scan. The random baseline wins by breaking the image, not by locating evidence better. We report this as a failure of the metric under volumetric masking, quantify the separation the methods do show under perturbation stability, where Grad-CAM reaches 0.924 and LRP 0.012 to 0.220, and argue that deletion protocols require manifold-preserving masking before they can adjudicate explanation quality in 3D medical imaging.

Index Terms—explainable AI, layer-wise relevance propagation, foundation models, self-supervised learning, uncertainty estimation, computed tomography

I. INTRODUCTION

Two claims are commonly made in medical imaging papers and rarely tested together. The first is that self-supervised pretraining on large unlabelled image collections improves downstream performance. The second is that a published attribution map explains the resulting model. Both are checkable, and checking them is the subject of this paper.

We fine-tune CT-FM [1], a SegResNet pretrained by contrastive self-supervised learning on 148,000 CT scans, for benign-versus-malignant lung-nodule classification on volumetric chest CT. We then explain it with layer-wise relevance propagation (LRP) [2] implemented in zennit [3], whose layer-type registry covers `Conv3d` and the 3D pooling operators and therefore applies to volumetric networks without modification,

and we benchmark LRP against Grad-CAM [4], the method most medical imaging work reaches for.

Our contributions are:

- 1) A label-efficiency ablation against an architecturally identical from-scratch control at 10%, 25% and 100% of labels, with a seed replicate that identifies which gains survive.
- 2) A faithfulness evaluation of three LRP composites and Grad-CAM on 3D CT, reporting the negative result that none beats random-order deletion, with a diagnosis of why.
- 3) A measurement of explanation stability under input perturbation, which separates the methods where deletion does not, and an analysis of whether faithfulness degrades with predictive uncertainty.

II. METHOD

A. Data

NoduleMNIST3D [8] comprises 3D chest CT patches centred on lung nodules drawn from LIDC-IDRI [9] and labelled benign or malignant. Splits are 1,158 training, 165 validation and 310 test volumes, with a 863:295 benign-to-malignant training imbalance. We therefore use class-weighted cross-entropy, report balanced accuracy and AUC alongside accuracy, and select models on validation AUC.

We use the 64³ release. This is not cosmetic: CT-FM downsamples by a factor of 16, so a 28³ volume is not divisible by the network stride and raises an error.

B. Model and three silent failure modes

CT-FM’s encoder produces a five-level feature pyramid; we take the bottleneck at 512 channels and 4³ resolution, apply global average pooling, LayerNorm and a linear head. Three implementation details are recorded because each fails quietly and would have produced a plausible but meaningless result.

First, the published checkpoint does not load into MONAI’s SegResNetDS. Parameter names differ, and `load_state_dict(strict=False)` matches *zero of*

TABLE I
LABEL-EFFICIENCY ABLATION, TEST AUC (SEED 0)

Labels	n	CT-FM	Scratch	Gain
10%	116	0.8283	0.8168	+0.0116
25%	290	0.8472	0.8124	+0.0348
100%	1,158	0.8951	0.8733	+0.0218

161 tensors without raising, so training proceeds on a randomly initialised network while the code appears correct.

Second, the from-scratch control must be architecturally identical. Constructing a second SegResNet from an inferred configuration yielded 19.9M parameters against the true 87.2M; that ablation would have measured capacity and attributed the difference to pretraining. Our control is the loaded model with every parameter reinitialised.

Third, pooling the full SegResNet output rather than the bottleneck gives the linear head two features. Combined with unnormalised activations, whose standard deviation is approximately 0.03, training loss remained at 6.37 with AUC 0.50.

The encoder is fine-tuned at 10^{-5} and the head at 10^{-3} : fine-tuning 87M pretrained parameters at the head’s rate displaces the representation that self-supervision produced, which is the quantity under test.

C. Attribution

LRP redistributes the output score backwards under conservation rules, producing a map at input resolution rather than at the resolution of the last convolutional feature map. We compare three composites, which encode different assumptions: ε -plus-flat, ε - γ -box with box bounds matching our $[-1, 1]$ inputs, and ε - $\alpha 2\beta 1$. Reporting one composite as “the LRP explanation” conceals how much the rule choice matters. Grad-CAM is computed on the encoder bottleneck and trilinearly upsampled.

D. Faithfulness, stability and uncertainty

Deletion progressively replaces the highest-ranked voxels with the dataset mean and tracks the target-class probability; a faithful map produces a steep early drop and hence a low area under the curve. **Insertion** restores highest-ranked voxels onto a blurred volume. **AOPC** is the mean probability drop across deletion steps. Every map is accompanied by a random-order baseline, because absolute values depend on model and data and only the gap against random is interpretable.

Stability is the mean correlation between the map on a clean input and on a noisy copy ($\sigma = 0.05$), noise the model itself ignores.

Uncertainty is estimated by MC dropout over 20 passes, giving predictive entropy and mutual information, plus expected calibration error.

III. RESULTS

A. Self-supervised pretraining helps, with one caveat

CT-FM initialisation wins at every budget (Table I, Fig. 1), reaching 0.895 AUC and 0.830 balanced accuracy at full supervision against 0.873 and 0.788 from scratch.

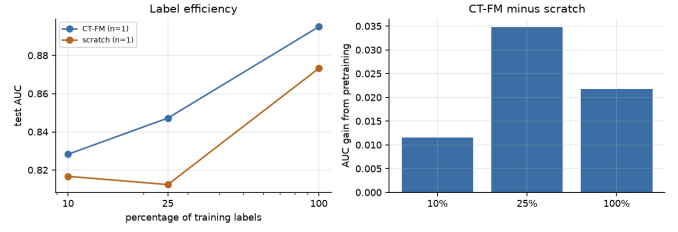


Fig. 1. Label efficiency and the AUC gain attributable to pretraining.

TABLE II
FAITHFULNESS AND STABILITY, 64 TEST VOLUMES

Method	Del.↓	Ins.↑	AOPC—rnd	Stab.
Random baseline	0.587	—	—	—
LRP ε +flat	0.819	0.972	−0.221	0.012
LRP $\varepsilon\gamma$ box	0.822	0.972	−0.224	−0.015
LRP $\alpha 2\beta 1$	0.816	0.973	−0.219	0.220
Grad-CAM	0.712	0.974	−0.119	0.924

The caveat is material and we state it rather than bury it. A seed replicate reversed the 10% result, giving scratch 0.805 against CT-FM 0.802. The smallest gain in Table I therefore lies inside seed variance. The 25% and 100% gains are larger and more plausible, but three or more seeds are required before any of these numbers should be quoted as established. Running the replicate is what exposed this, and it cost fourteen minutes.

B. Calibration

Expected calibration error under MC dropout is 0.145 with mean predictive entropy 0.085: the model is confident and less accurate than its confidence implies, the usual direction for a small fine-tuned network on an imbalanced task.

C. No attribution method beats random deletion

A faithful map should sit *below* the random baseline on deletion AUC. All four sit well above it (Table II, Fig. 2): removing the voxels these methods rank highest damages the prediction *less* than removing random voxels.

We do not read this as evidence that the maps are worthless. It is a property of the metric under volumetric masking. Random deletion scatters replaced voxels uniformly through the volume; at 5% this already produces something no CT scanner would output, and the network’s response collapses because the input has left the data manifold. Attribution-guided deletion removes a compact contiguous region, which remains a far more plausible scan. The random baseline wins by destroying the image, not by identifying evidence.

Two consequences follow. On this task and at this scale, deletion is not a usable adjudicator of explanation quality, and a manifold-preserving masking scheme, such as inpainting the deleted region, is required before the protocol can discriminate. Insertion, meanwhile, separates nothing: all four methods recover to 0.972–0.974.

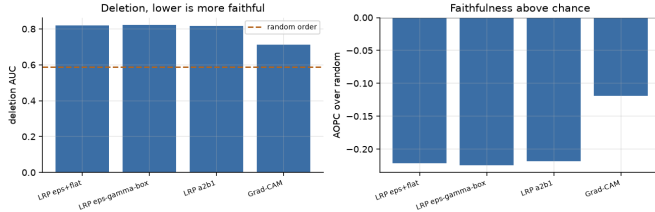


Fig. 2. Deletion AUC against the random-order baseline, and AOPC relative to random. All four attribution methods fall on the wrong side of the baseline.

D. Stability separates what deletion cannot

Under perturbation the methods diverge sharply. Grad-CAM maps are almost invariant (0.924) while LRP maps essentially decorrelate (0.012, -0.015 , 0.220). This is not straightforwardly a defect of LRP. Grad-CAM is stable *because* it is coarse: a 4^3 map upsampled to 64^3 has little freedom to move. LRP resolves to input resolution and pays in sensitivity. Whether a fine-grained map that shifts under imperceptible noise is preferable to a coarse one that does not is a genuine question that neither metric here settles.

E. Uncertainty against faithfulness

Spearman correlations between predictive entropy and per-sample AOPC are $+0.190$, $+0.174$ and $+0.221$ for the three LRP composites and -0.124 for Grad-CAM. The LRP composites agree with each other and lean weakly positive, meaning explanations are if anything marginally more faithful where the model is least certain. With 64 volumes none of these is significant; the honest reading is that the experiment is underpowered rather than that the effect is absent. The question, posed for segmentation by Salahuddin et al. [6], is the right one and needs an order of magnitude more data.

IV. LIMITATIONS

The headline ablation uses a single seed, and the one replicate we ran already overturned its smallest gain. The explanation analysis covers 64 volumes, sufficient to demonstrate that the deletion metric misbehaves but not to resolve a weak correlation. Our deletion baseline replaces voxels with the dataset mean; a blurred or inpainted baseline stays nearer the data manifold and would likely change the conclusion of Section III-C, and this is the first thing we would fix. Inputs are 64^3 patches rather than whole chest CT, so a model pretrained on complete scans may not be exercised as intended. Uncertainty uses MC dropout only; deep ensembles are implemented but their members were not trained in time.

V. CONCLUSION

Self-supervised CT pretraining improved downstream nodule classification at every label budget, with the smallest gain inside seed noise. More usefully, the standard deletion protocol ranked every attribution method below a random baseline, and the cause is the baseline’s distribution shift rather than the explanations’ quality. Faithfulness metrics inherited from 2D natural images require manifold-preserving masking

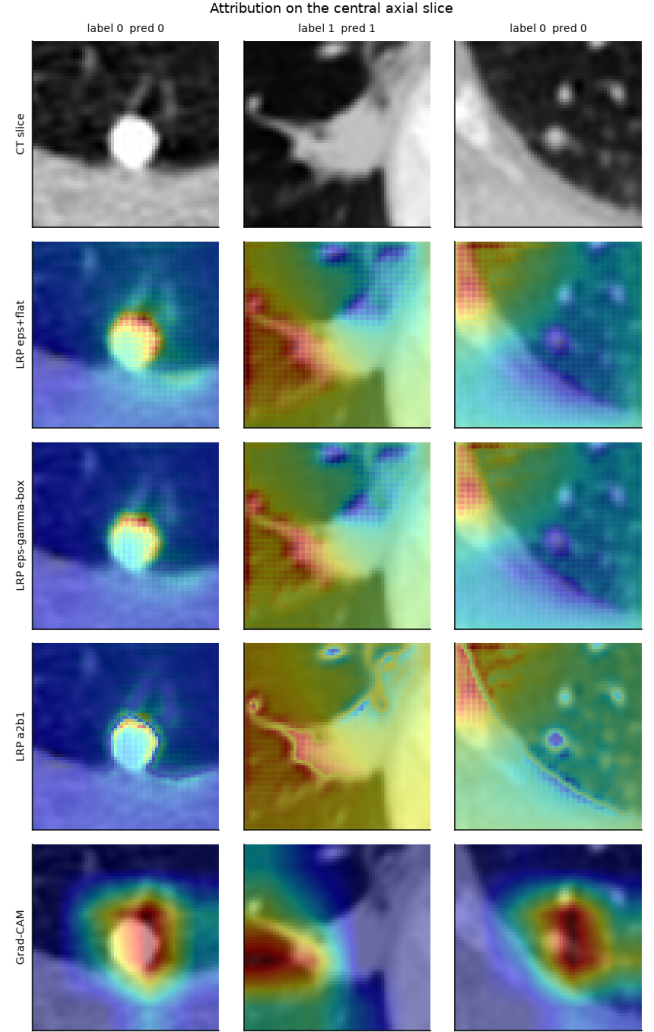


Fig. 3. Attribution on the central axial slice for three test volumes.

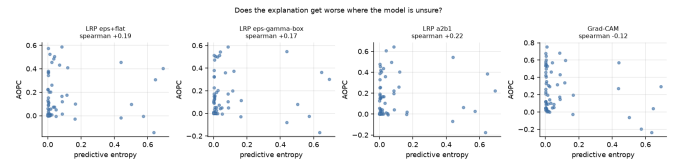


Fig. 4. Per-sample predictive entropy against AOPC. Correlations are weak and the three LRP composites agree with one another and disagree with Grad-CAM.

before they can be trusted to adjudicate explanation quality on volumetric medical data. We recommend reporting the random control explicitly, since without it the same numbers would have read as a straightforward endorsement of whichever method scored best.

REPRODUCIBILITY

Code, result files and all figures are available at <https://github.com/Joana-Mansa/ct-fm-lrp-uncertainty>. The full ablation runs in approximately 14 minutes on one A100. On Volta

hardware, `torch` must be pinned to 2.5.1+cu121; releases from 2.14 ship an architecture list beginning at `sm_75` and every CUDA call fails.

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